End-of-year and oral workshops

O1 · Turning a topic into a question

S82 · 2 h. Choose a situation that allows reasoning from the syllabus, then write a specific question. Examples: "How do we choose a number of trials to make failure rare?"; " How does a cooling model predict a waiting time?"; "Why does a mean become more stable as the sample grows?" Produce a page containing the question, the assumptions, a numerical example, an accessible proof and a limitation of the model. Allow 20 min of individual research, 30 min of mathematical work, 30 min of discussion in groups, 25 min of writing and 15 min of feedback.

O2 · Presenting and answering

S84 · 2 h. Prepare a 10-minute presentation and answers to the following questions: why this model ? which assumption is decisive? what does each variable mean? which result is proved? what changes for another value of the parameter? Practise in groups of three: a speaker, a listener and an observer. The observer notes one clear step, one step needing justification and one wording to improve. Swap roles. A video is only useful if it is permitted and actually watched again.

O3 · Mock oral exam

S87 and S93. Prepare two questions and give a presentation followed by questioning. Use a simple grid: mathematical accuracy, structure, explanation of notation, response to objections, quality of the oral exchange. The final arrangements are those given in the instructions and notices for the exam session.